sEMG Muscle Activation Report

Participant P6 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Sustained
Consistency	Consistent
Dominant Sensors	S4
Weakest Sensors	S2
Assessment Confidence	0.95

Clinical Summary: The session demonstrates a stable, high-intensity sustained contraction with significant co-contraction between flexor and extensor groups. S4 is the primary driver of the gesture. S8 shows a notable decrease in amplitude over time, while S6 shows a compensatory ramp-up, suggesting potential localized fatigue or shift in motor control strategy.

Key Findings:
- High co-contraction ratio
- S8 ramp-down profile
- S6 ramp-up profile

Session Overview

Session Duration	1202.4 s (20.0 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.019991 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.012043	0.220	120.1	102.1	1448	Moderate	Sustained
S2	0.008446	0.096	108.0	88.0	330	Moderate	Sustained
S3	0.018561	0.157	99.8	87.7	3026	High	Sustained
S4	0.040835	0.197	111.5	99.2	10311	Very High	Sustained
S5	0.027778	0.162	137.6	126.9	4985	Very High	Sustained
S6	0.011569	0.246	97.2	68.8	975	Moderate	Ramp-up
S7	0.014804	0.129	99.7	64.0	1836	High	Sustained
S8	0.025892	0.346	66.7	51.5	2195	Very High	Ramp-down

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S4: Highest WAMP/RMS

S6: Increasing activation trend

S8: Decreasing activation trend

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S4	Primary contributor
Weakest Sensor	S2	Lowest activity
Dominant-to-Weakest Ratio	4.83×	Activation disparity
Concentration Index	0.162	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.996	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.222	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.855	Agonist/antagonist balance

Flexor/Extensor Balance	1.02	>1 = flexor-dominant
Overall Temporal Profile	Sustained	Session-wide activation trend

Co-contraction: Detected — Agonist and antagonist muscles activating simultaneously (normal for power grip).

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor (S1-S4) and Extensor (S5-S8) groups show balanced activation (ratio 1.0187), consistent with co-contraction during power grip.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S4, S5

Evidence: S4 shows highest WAMP (10310.7) and S5 shows high WAMP (4984.9), indicating significant motor unit recruitment during sustained contraction.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.